PROJECT SYNOPSIS

Title of the Project

AI-Based Social Issue Detection from Social Media

Team members

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1. Introduction

In recent years, social media platforms have become a common medium for people to express their thoughts, complaints, emotions, and daily problems. Many individuals share serious concerns such as harassment, depression, cyberbullying, public safety issues, and civic problems online before reporting them through official channels. However, due to the enormous volume of posts generated every day, it is impossible to manually monitor and identify genuine and urgent issues.

This project proposes a web-based analytical system that uses Artificial Intelligence and Natural Language Processing (NLP) to analyze publicly available textual posts. The system identifies whether a post represents a real issue, classifies its type, and estimates its severity level. The goal is to demonstrate how technology can assist in early identification and prioritization of important social and psychological problems.

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2. Problem Statement

- Large amounts of online data make manual monitoring difficult
- Genuine problems remain unnoticed among jokes, spam, and casual posts
- No automated system exists to classify and prioritize issues based on seriousness

- Authorities and organizations cannot easily identify urgent cases in time
- Early warning signs such as mental distress or threats are often ignored

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3. Objectives of the Project

The main objectives of this project are:

To develop a web-based platform for analyzing textual posts

To detect genuine social and psychological issues automatically

To classify problems into categories (social, civic, psychological, crime, emergency)

To estimate the severity level (Low, Medium, High, Critical)

To highlight high-risk issues through a dashboard

To demonstrate how NLP can convert unstructured text into meaningful insights

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4. Scope of the Project

The system will be useful for:

- **Academic research and demonstration**
- **Social awareness analysis**
- **Understanding public concerns**

The project will work on a prepared dataset of public posts and will not collect private user data or connect to real government systems. It is a research prototype intended for analysis and educational purposes.

Future enhancements may include:

- **Real-time data analysis**
- **Multi-language support**
- **Integration with support organizations**

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5. Proposed System

User Roles

User Role

Admin (Researcher)

- Upload dataset
- Run analysis
- View results and reports

Key Functionalities

1. Upload textual dataset
2. Store posts in database
3. Process text using NLP
4. Identify genuine issues
5. Classify issue category
6. Assign risk severity level
7. Display results on dashboard
8. Highlight critical cases

6. AI & Machine Learning Features

The system uses Natural Language Processing to:

- Clean and preprocess text
- Detect emotional and distress language
- Classify the type of problem
- Extract important keywords
- Estimate seriousness of the issue

The model analyzes linguistic patterns rather than verifying factual truth. It provides a probability-based assessment to assist prioritization**.**

7. System Architecture

Dataset Input → Backend Server → NLP Processing → Database → Dashboard Output

1. Admin uploads dataset
2. Backend sends text to AI model
3. Model analyzes and classifies posts
4. Results stored in database
5. Dashboard displays categorized issues

8. Technology Stack

Frontend:
HTML, CSS, JavaScript, Bootstrap/React

Backend:
Node.js, Express.js

Database:
MongoDB

AI/NLP:
Python, NLP libraries (Scikit-Learn / NLTK)

Tools:
VS Code, Git & GitHub

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9. Dataset Description

The dataset will include:

- Property location
- Distance from college
- Rent amount
- Facilities
- Room type
- User preference data

Data can be:

- Collected manually
- Dummy dataset for mini project
- Enhanced using real listings

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10. Methodology

1. Data collection
2. Data preprocessing
3. Feature selection
4. ML model training
5. Backend integration
6. Frontend development
7. Testing and validation
8. Deployment (optional)

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11. Expected Outcomes

- Automatic detection of issues
- Categorized problem identification
- Risk severity assessment
- Visual dashboard highlighting critical cases
- Demonstration of AI-based social analysis

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12. Advantages of the System

- Early identification of serious issues
- Automated filtering of large data
- Structured understanding of public concerns
- Useful academic research demonstration

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13. Limitations

- Accuracy depends on dataset quality
- Works only on textual input
- Does not verify real-world events (analysis only)

14. Future Enhancements

- Real-time social media integration
- Multi-language detection
- Advanced machine learning models
- Support organization integration

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15. Conclusion

The proposed system demonstrates how Artificial Intelligence and Natural Language Processing can be used to analyze online textual data and identify important social and psychological issues. By classifying and prioritizing posts based on severity, the platform provides an early-warning analytical approach that can help understand public concerns more efficiently. The project serves as a research-oriented prototype showing the practical application of AI in social problem analysis.